Hermes Agent × Raspberry Pi 5 × 1996 Toyota Camry
Complete Integration Guide

A comprehensive technical reference for integrating a Raspberry Pi 5 into a 1996 Toyota Camry platform

Version 1.0 | June 2026
150 Integrations | Wiring Diagrams | GPIO Reference | Power Architecture

Table of Contents

1. Parts List

Part Name	Model / Part Number	Purpose	Price Est.	Supplier
Raspberry Pi 5	SC1111 / 8GB RAM	Main compute unit	$80	PiShop.us / Adafruit
Pi 5 Active Cooler	SC1148	Thermal management	$5	PiShop.us
M.2 NVMe HAT	Pineberry Pi HatDrive! Top / Waveshare	NVMe SSD interface	$25	Pineberry Pi / Amazon
NVMe SSD	Samsung 990 EVO 1TB	Mass storage (video/logs)	$90	Amazon / Newegg
12V→5V Buck Converter	LM2596S / XL4015 module	Power supply (12V→5V/5A)	$8	Amazon / AliExpress
Dual Camera Module	IMX219 8MP ×2 (or IMX708)	Front & rear video capture	$60	Arducam / PiShop
Camera Cable Ext.	CSI-2 22-pin 1m ribbon	Camera routing in vehicle	$12	Arducam
OBD-II Adapter	ELM327 USB / STN1110	Vehicle diagnostic interface	$15	Amazon
GPS Module	u-blox NEO-M9N / USB GPS	Position & speed logging	$35	SparkFun / Adafruit
4G/LTE Modem	Quectel EC25 / SIM7600G-H	Cellular connectivity	$45	Sixfab / Waveshare
ADS-B Receiver	RTL-SDR Blog v4	Air traffic monitoring	$35	RTL-SDR.com
TPMS Reader	RTL-SDR + rtl_433	Tire pressure monitoring	$35	RTL-SDR.com
USB Audio Interface	Behringer UCA222 / CM108	Aux audio in/out	$30	Amazon / Sweetwater

Relay Board	5V 4-channel relay module	Ignition / accessory control	$6	Amazon
Logic Level Shifter	TXS0108E / BSS138	3.3V↔5V signal translation	$4	Adafruit / Amazon
Fuse Tap Set	ATO/ATC add-a-circuit	Clean 12V tap from fusebox	$8	AutoZone / Amazon
Wiring Kit	18 AWG primary wire 100ft	Power & signal distribution	$20	Amazon
Project Enclosure	ABS plastic 200×150×75mm	Pi housing under dash	$15	Amazon / Hammond
MicroSD Card	SanDisk Extreme 64GB	Boot / OS	$12	Amazon
USB Hub	Powered 7-port USB 3.0	Peripheral expansion	$18	Amazon
Accelerometer	MPU-6050 / LIS3DH (I2C)	G-force / crash detection	$5	Adafruit / Amazon
Temperature Sensor	BME280 (I2C)	Cabin / ambient temp/humidity	$8	Adafruit
LED Strip	WS2812B 5V 1m 60 LEDs	Ambient / status lighting	$10	Amazon
CAN Bus Interface	MCP2515 + TJA1050 (SPI)	Vehicle network (if equipped)	$12	Amazon / Seeed
USB-C Power Cable	Right-angle 1.5m 5A	Pi 5 power delivery	$8	Amazon
SIM Card	IoT data plan SIM	Cellular data	$10/mo	Hologram / Twilio

> **Note:** The 1996 Toyota Camry uses OBD-II (mandated from 1996 in the US). CAN bus may not be present on all 1996 models; verify under-dash connector pinout before purchasing CAN hardware.

2. GPIO Pinout Reference — Raspberry Pi 5

The Pi 5 40-pin header maintains backward compatibility with Pi 4 but adds a dedicated PCIe / power controller (RP1). Key pins for this integration:

Pin	Name	Function	Used For
1	3.3V	Power	Sensors, level shifter VCCA
2	5V	Power	Relays, LED strip (via level shifter)
3	GPIO2	SDA1 (I2C)	I2C bus — BME280, MPU-6050, OLED
4	5V	Power	—
5	GPIO3	SCL1 (I2C)	I2C bus clock
6	GND	Ground	Common ground
7	GPIO4	GPCLK0	1-Wire (optional)
8	GPIO14	TXD0 (UART)	GPS / LTE AT console
9	GND	Ground	—
10	GPIO15	RXD0 (UART)	GPS / LTE AT console
11	GPIO17	GPIO	Relay CH1 (ignition sense)
12	GPIO18	PWM0	WS2812B LED strip data
13	GPIO27	GPIO	Relay CH2 (accessory cut)
14	GND	Ground	—
15	GPIO22	GPIO	Relay CH3 (camera power)
16	GPIO23	GPIO	Relay CH4 (LTE modem power)
17	3.3V	Power	—
18	GPIO24	GPIO	Alarm / buzzer output
19	GPIO10	MOSI (SPI)	SPI devices / CAN MCP2515

20	GND	Ground	—
21	GPIO9	MISO (SPI)	SPI devices / CAN MCP2515
22	GPIO25	GPIO	Status LED / heartbeat
23	GPIO11	SCLK (SPI)	SPI clock
24	GPIO8	CE0 (SPI)	CAN bus chip select
25	GND	Ground	—
26	GPIO7	CE1 (SPI)	Reserve SPI
27	ID_SD	I2C ID	HAT EEPROM
28	ID_SC	I2C ID	HAT EEPROM
29	GPIO5	GPIO	PIR / motion sensor
30	GND	Ground	—
31	GPIO6	GPIO	Door lock sense
32	GPIO12	PWM1	Fan PWM control
33	GPIO13	PWM1	Reserve PWM
34	GND	Ground	—
35	GPIO19	SPI1_MISO	Reserve
36	GPIO16	SPI1_CE2	Reserve
37	GPIO26	GPIO	Reserve
38	GPIO20	SPI1_MOSI	Reserve
39	GND	Ground	—

40	GPIO21	SPI1_SCLK	Reserve

> **Pi 5 specific:** UART now routes through RP1; enable via `raspi-config` or `/boot/firmware/config.txt`. PCIe M.2 uses the new FPC connector (not GPIO pins).

3. Power Wiring Architecture

3.1 12V→5V Buck Converter

```
Camry Battery (+)  ────┬── Fuse 10A ───→  Buck IN+  ───→  5V/5A OUT+  ──→  Pi 5 USB-C VBUS
(Terminal / fusebox)   │                                   │
                       │                                   ├──→  USB Hub VBUS
                       │                                   ├──→  Relay board VCC
                       │                                   ├──→  LTE modem (via relay)
                       │                                   └──→  Camera modules (via relay)
Camry Chassis GND  ────┴──────────────────────────────────→  Buck GND / Pi GND
```

3.2 Ignition-Switched vs Always-On Circuits

Circuit	Source	Fuse	Purpose
Always-On 12V	Battery + / Fusebox BAT	15A	Buck converter input, LTE modem (standby), GPS backup
Ignition-Switched 12V	IG1 / ACC fuse tap	10A	Pi main power, cameras, USB hub, relays
Accessory 12V	ACC circuit	5A	LED strip, display backlight

3.3 Relay Logic

```
Ignition Sense (GPIO17) ──→  Opto-isolator ──→  Relay CH1
Relay CH1 (NC/NO) ──→  Controls 12V feed to Buck IN+ (or Pi 5V enable)

Software watchdog:
  - If ignition OFF & no activity for 10 min → GPIO27 triggers Relay CH2
  - Relay CH2 cuts USB hub & camera power
  - Pi enters low-power mode (rtcwake or systemd suspend)
  - LTE modem stays on minimal heartbeat (if configured)
```

> **Warning:** Always fuse both +12V lines within 150mm of the battery or fusebox tap. Use automotive
> ATO/ATC fuses rated for continuous load. The buck converter MUST be rated for automotive transient

> suppression (load dump / ISO 7637-2 pulse 5a). Add a TVS diode (SMBJ24A) across buck input.

4. M.2 NVMe Storage Architecture

The Pi 5 exposes a single-lane PCIe 2.0 interface via a dedicated FPC connector. An M.2 HAT (or HAT+ adapter) converts this to an M.2 2230/2242/2280 slot.

4.1 Physical Topology

```
Pi 5 SoC (BCM2712)
     │
     ├── PCIe x1 Gen2 (5 GT/s) via FPC
     │
     └── M.2 HAT (RP1-managed power sequencing)
              │
              ├── M.2 Key-M slot (NVMe SSD)
              │       └── Samsung 990 EVO 1TB (or WD SN770)
              │
              └── Jumper: enable/disable SSD power (for safe shutdown)
```

4.2 Storage Partitioning Strategy

Partition	Size	Mount	Purpose
/dev/nvme0n1p1	500MB	/boot/firmware	Bootloader + kernel (optional, can boot from SD)
/dev/nvme0n1p2	32GB	/	Root filesystem (OS + apps)
/dev/nvme0n1p3	64GB	/var/log	Structured logs (OBD, GPS, accelerometer)
/dev/nvme0n1p4	Remaining	/mnt/video	Rolling video buffer + encrypted archive

4.3 Enable NVMe Boot

```
# /boot/firmware/config.txt
[all]
dtparam=pciex1
dtparam=nvme
```

> **Note:** As of mid-2024 firmware, Pi 5 supports NVMe boot. Update bootloader first: `sudo rpi-eeprom-update -a`. Use `lsnvme` or `ls /dev/nvme*` to verify detection.

5. Video Recording Architecture

5.1 Dual Camera Setup

```
   Pi 5 CSI-2 port 0 (CAM0) ──→  22-pin FPC extender ──→  Front camera (IMX219)
        │                                                    Windshield mount
        │                                                    1080p30 / 720p60
        │
        └── CSI-2 port 1 (CAM1) ──→  22-pin FPC extender ──→  Rear camera (IMX219)
                                                               Rear window mount
                                                               1080p30 / 720p60
```

5.2 Rolling Buffer & Encryption Pipeline

```
   Camera 0/1 (YUV/RGB) ──→  libcamera-vid / rpicam-vid  ──→  H.264/H.265 encoder (hardware)
                                                                    │
                                                                    ├──→  RAM ring buffer (5 min, ~1.5GB)
                                                                    │       └── Emergency save on G-
shock / GPIO trigger
                                                                    │
                                                                    └──→  NVMe partition /mnt/video
                                                                                │
                                                                                ├──→  Hourly segments
(MP4/MKV)
                                                                                │       └── Rotation: keep
last 48h
                                                                                │
                                                                                └──→  Encrypted archive
(LUKS or gocryptfs)
                                                                                        └── Long-term
incidents (passphrase or keyfile)
```

5.3 Software Stack

Component	Package / Tool	Role
Camera driver	libcamera / rpicam-apps	Capture & tuning
Encoder	FFmpeg (V4L2 M2M)	Hardware H.264 encode
Segmenter	FFmpeg -f segment	Split by time

Buffer manager	tmpfs + inotify	RAM ring buffer
Encryption	gocryptfs / LUKS	At-rest protection
Trigger	MPU-6050 + GPIO	Crash / motion detect

> **Tip:** For a 1996 Camry without built-in camera power lines, run fused 5V from the buck converter directly to camera modules. Keep CSI-2 ribbon cables away from ignition coil wiring to reduce EMI.

6. Integration Categories (150 Total)

6.1 Engine (15 integrations)

E01 — OBD-II Live Data Stream

Read RPM, MAP, coolant temp, throttle position, fuel trim via ELM327 USB. Poll at 10Hz. Log to SQLite.

E02 — Check Engine Light Decoder

Parse DTCs (P0xxx, B0xxx). Cross-reference with Toyota service manual. Display plain-text fault on OLED.

E03 — Fuel Economy Calculator

Compute MPG from MAF or fuel injector duty cycle (if available). Store per-trip averages.

E04 — Engine Warm-Up Timer

Track coolant temp rise from cold start. Alert when optimal temp reached.

E05 — Knock Sensor Monitor

If OBD exposes knock retard, log events. Correlate with accelerometer data.

E06 — Battery Health Logger

Log voltage (via INA219 on 12V rail) and crank voltage dip. Predict battery replacement.

E07 — Alternator Output Monitor

Monitor 14V rail stability. Alert on under/over-voltage conditions.

E08 — Oil Pressure Proxy

1996 Camry oil pressure switch is binary. Log open/closed state; infer pressure from RPM/load model.

E09 — Exhaust Temp Estimator

Model EGT from OBD2 coolant temp, intake air temp, and fuel trim.

E10 — Ignition Timing Logger

Log spark advance from OBD. Detect knock-related retard patterns.

E11 — Idle Speed Tracker

Monitor idle RPM stability. Detect vacuum leaks or IAC issues.

E12 — Air/Fuel Ratio Display

Display calculated AFR from narrowband O2 sensor (via OBD) or wideband if added.

E13 — EGR Flow Monitor

Log EGR commanded vs actual (if supported by ECU).

E14 — Transmission Shift Logger

Log gear position (A340E solenoid states via OBD if exposed). Count shifts per trip.

E15 — Predictive Maintenance Score

ML model (scikit-learn on Pi) scoring engine health from multi-sensor trends.

6.2 Safety (12 integrations)

S01 — Forward Collision Warning (Camera)

OpenCV-based vehicle detection on front camera. Audio alert if TTC < 2.5s.

S02 — Lane Departure Warning

Hough transform lane detection. Alert on unintended lane change (no turn signal).

S03 — Rear Cross-Traffic Alert

Rear camera motion detection when in reverse (detected via OBD gear or backup light tap).

S04 — Blind Spot Monitor (Camera)

Side mirror mounted USB camera (optional). Optical flow object tracking.

S05 — G-Force / Crash Detection

MPU-6050 at 1kHz. Trigger emergency video save + SMS alert on >3G impact.

S06 — Driver Drowsiness Detection

USB cabin camera + face landmarks. Alert on prolonged eye closure.

S07 — Speed Limit Overlay

GPS + OpenStreetMap speed limit data. HUD-style audio warning on exceed.

S08 — Emergency Brake Light Flash

Rapid deceleration (>0.4G) triggers relay to flash brake lights (where legal).

S09 — Tire Pressure Monitor (RTL-SDR)

rtl_433 reads 315/433MHz TPMS sensors. Display pressure & temp per tire.

S10 — Flood / Water Ingress Sensor

Capacitive water sensor in spare tire well. Alert if water detected.

S11 — Post-Crash Auto-Notify

On crash detect: LTE modem sends GPS coordinates + timestamp to emergency contact.

S12 — Backup Battery Monitor

Monitor backup 12V battery (if installed) for dashcam continuity during main battery disconnect.

6.3 Navigation (10 integrations)

N01 — GPS Position Logging

u-blox NEO-M9N at 10Hz. Log to GPX for trip replay.

N02 — Offline Maps (OSM)

Pre-rendered OSM tiles on NVMe. Navit or OSMAnd-like renderer.

N03 — Turn-by-Turn Voice

Open-source routing engine (OSRM/Valhalla). TTS via espeak or Piper.

N04 — Traffic Overlay (LTE)

Fetch live traffic via LTE. Overlay on offline map.

N05 — POI Database

Custom POI: gas stations, Toyota dealers, rest stops. SQLite geospatial index.

N06 — Route Efficiency Scorer

Score routes by fuel economy history on that road segment.

N07 — Parking Spot Logger

Auto-save GPS on ignition off. Navigate back to car.

N08 — Elevation Profile

SRTM elevation data. Display grade and predicted fuel impact.

N09 — Compass / Heading

GPS course + magnetometer (QMC5883). Display on OLED or web UI.

N10 — Geo-Fencing Alerts

Alert when vehicle enters/exits defined zones (home, work, restricted).

6.4 Infotainment (12 integrations)

I01 — Local Music Server

MPD (Music Player Daemon) serving FLAC/MP3 from NVMe. Web control via Rompr.

I02 — FM Radio (RTL-SDR)

rtl_fm for FM demod. Software RDS decoder for station info.

I03 — Bluetooth Audio (A2DP)

Pi 5 built-in Bluetooth + PulseAudio. Stream from phone.

I04 — Podcast Downloader

gPodder / cron + wget. Auto-download over WiFi (home) or LTE.

I05 — Web Dashboard

Flask/FastAPI served on localhost. Displayed via 7" DSI touchscreen or tablet.

I06 — Voice Assistant (Local)

Whisper (STT) + Piper (TTS) + local LLM (llama.cpp). No cloud required.

I07 — Audio EQ / DSP

CamillaDSP for FIR/IIR filtering. Room correction via mic (optional).

I08 — Video Playback

Kodi or VLC. Only when parked (ignition OFF + handbrake sense).

I09 — USB Media Import

Auto-mount USB → rsync to NVMe music library on insert.

I10 — Backup Camera Auto-Display

Detect reverse via OBD or backup light tap → switch display to rear camera.

I11 — OBD Data HUD

Real-time dash: RPM, speed, coolant temp, MPG on touchscreen.

I12 — Steering Wheel Controls

Resistive ladder from factory wheel → ADC (MCP3008) → volume/skip commands.

6.5 Climate (8 integrations)

C01 — Cabin Temperature Monitor

BME280 inside dash. Log temp/humidity. Detect HVAC effectiveness.

C02 — Ambient Temperature Display

External BME280 in front grille (weatherproofed). Display on dash.

C03 — HVAC Status Logger

Tap AC clutch wire + blend door servo PWM (if accessible). Log on/off states.

C04 — Pre-Condition Timer

Scheduled engine start not recommended (safety). Instead: alert if cabin >90°F on parked car.

C05 — Air Quality (VOC/CO2)

SGP30 or SCD40 sensor. Alert on high VOC; suggest recirculate.

C06 — Solar Load Estimator

Use GPS time + heading + external temp to model cabin solar heating.

C07 — Defrost Reminder

If ambient <5°C and humidity >85%, remind driver to enable defrost.

C08 — Cabin Air Filter Timer

Track HVAC hours. Remind replacement at 200h (Toyota spec).

6.6 Security (12 integrations)

SEC01 — GPS Tracker / Anti-Theft

LTE modem reports GPS every 60s when ignition off. Geofence breach alert.

SEC02 — Motion-Activated Recording

PIR or camera motion detection when parked. Save to encrypted partition.

SEC03 — Remote Lock/Unlock (via SMS)

SMS command to LTE modem → GPIO relay to door lock actuator (parallel to factory).

SEC04 — Ignition Kill Switch

Relay in starter circuit. Disable via authenticated SMS/app.

SEC05 — Alarm Siren Integration

Tap into factory alarm horn or add standalone siren. Trigger on unauthorized entry.

SEC06 — Glass Break Detection

Audio FFT analysis on USB mic. Detect glass break frequency signature.

SEC07 — Key Fob Signal Logger

RTL-SDR logs 315MHz key fob transmissions. Detect replay attacks.

SEC08 — Cabin Occupancy Detection

PIR + ultrasonic. Alert if child/pet left in parked car (temp > threshold).

SEC09 — Tamper Detection

Case open switch + accelerometer. Alert if Pi enclosure accessed.

SEC10 — Remote Engine Start (with caveats)

SMS triggers starter relay. Only with manual transmission interlock and brake sense.

SEC11 — Trunk Open Alert

Tap trunk latch switch. Alert if left open >5 min.

SEC12 — Valet Mode

Geofence + speed limiter logging. Notify owner if valet exceeds boundaries.

6.7 Lighting (10 integrations)

L01 — Ambient LED Strip Control

WS2812B under dash / footwells. Color temp matches time of day.

L02 — Headlight Auto-On

Photoresistor + GPS sunset data. Auto-enable headlights via relay if driver forgets.

L03 — Brake Light Strobe

Rapid flash on hard braking (relay tapped into brake switch wire).

L04 — Turn Signal Monitor

Log turn signal usage. Alert if left on >2 min (lane change vs turn).

L05 — Interior Light Delay

Extend dome light timeout after ignition off (relay controlled).

L06 — License Plate Illumination Check

Current sense on plate light circuit. Alert if bulb out.

L07 — Fog Light Logic

Prevent fog light use when high beams on (compliance). Log usage.

L08 — Welcome / Puddle Lights

Unlock signal → fade-on footwell LEDs for 30s.

L09 — Reverse Light Current Monitor

INA219 on reverse circuit. Detect bulb failure or trailer load.

L10 — Daytime Running Light Dimming

PWM dim DRLs (if LED retrofitted) when turn signal active.

6.8 Data Logging (15 integrations)

DL01 — Trip Computer

Distance, time, avg speed, fuel used, cost per trip. SQLite backend.

DL02 — OBD2 Data Historian

Time-series DB (InfluxDB or TimescaleDB). Retention: 2 years.

DL03 — GPS Track Archive

GPX + GeoJSON export. Search by date, location, or event.

DL04 — Video Event Archive

Encrypted storage of flagged events (crash, horn, manual trigger).

DL05 — Accelerometer Black Box

1kHz XYZ log for 30s pre/post crash. Circular buffer in RAM.

DL06 — CAN Bus Logger (if equipped)

MCP2515 dumps all frames to candump format. Parse with python-can.

DL07 — Fuel Purchase Logger

Manual entry via web UI. Compute price trends and MPG per fill-up.

DL08 — Maintenance Schedule Tracker

Mileage/time-based reminders: oil, tires, belts, coolant.

DL09 — Driving Behavior Score

Score: harsh accel/brake/cornering. Gamify efficient driving.

DL10 — Emissions Readiness Monitor

Track OBD readiness bits. Alert before smog test if not ready.

DL11 — Weather Correlation

Fetch weather API data per trip. Correlate temp/humidity with MPG.

DL12 — Tire Wear Estimator

Log mileage + alignment G-forces. Estimate tread depth over time.

DL13 — Battery Cycle Counter

Count deep discharges. Predict replacement at 80% capacity.

DL14 — Oil Change Interval Optimizer

Adjust interval based on driving style (short trips vs highway).

DL15 — Remote Data Sync

rsync / rclone to home NAS or cloud (encrypted) when on home WiFi.

6.9 AI / LLM (10 integrations)

AI01 — Local LLM (llama.cpp)

Run quantized LLM (Llama 3 8B Q4) on Pi 5 (8GB). ~4 tok/s. Offline capable.

AI02 — Voice Assistant (Hermes Agent)

Wake word → Whisper STT → local LLM → Piper TTS. Full pipeline on-device.

AI03 — OBD Fault Code Explainer

LLM receives DTC + live data. Explains likely cause and fix in plain English.

AI04 — Predictive Maintenance (ML)

Scikit-learn regression on sensor trends. Predict component failure 500mi ahead.

AI05 — License Plate Reader (ANPR)

OpenALPR or custom YOLO+OCR on rear camera. Log plates in proximity.

AI06 — Object Detection (Coral TPU)

USB Coral Edge TPU for 30fps MobileNet SSD. Pedestrian/cyclist alert.

AI07 — Natural Language Vehicle Queries

"What's my coolant temp?" → LLM queries DB → spoken response.

AI08 — Summarize Drive Logs

LLM summarizes daily trips, anomalies, and recommendations.

AI09 — Sentiment Analysis (Cabin Audio)

Detect driver stress from voice tone (optional, privacy-sensitive).

AI10 — Image Captioning (Cameras)

BLIP / LLaVA on snapshots. "Describe what the front camera sees."

6.10 Power Management (10 integrations)

PM01 — Smart Shutdown Controller

Ignition OFF → 10 min timer → graceful shutdown → relay cuts 5V.

PM02 — Deep Sleep / RTC Wake

Pi 5 suspend-to-RAM. RTC alarm wakes for scheduled tasks.

PM03 — Battery Voltage Logger

INA219 on 12V rail. Log every 10s. Plot voltage curves.

PM04 — Solar Panel Input (Optional)

20W solar on rear deck → MPPT controller → trickle charge battery.

PM05 — Load Shedding

If battery <12.0V, sequentially cut: cameras → LTE → Pi (hibernate).

PM06 — Power Budget Calculator

Real-time sum of all loads. Alert if exceeding alternator capacity at idle.

PM07 — USB Port Power Control

Per-port power switching via smart hub or GPIO-controlled MOSFETs.

PM08 — Alternator Load Balancer

Defer high-power tasks (video encode, LLM inference) until RPM >1500.

PM09 — Supercapacitor UPS (Optional)

10F supercap bank provides 10s ride-through for safe shutdown.

PM10 — Thermal Throttling Monitor

Log CPU temp. Reduce load if >80°C. Control fan PWM via GPIO12.

6.11 M.2 / Video (8 integrations)

MV01 — NVMe Boot

Boot Pi 5 directly from NVMe SSD. Faster I/O, larger storage.

MV02 — Dual CSI Camera Capture

Simultaneous 1080p30 front + rear via libcamera. HW encode to H.264.

MV03 — Rolling Buffer Manager

tmpfs ring buffer (5 min). Auto-save on trigger. Offload to NVMe.

MV04 — Encrypted Video Vault

gocryptfs on /mnt/video. Keyfile on USB or passphrase on boot.

MV05 — HDMI Output (Dash Display)

Pi 5 dual HDMI: one for dash touchscreen, one for rear passenger.

MV06 — DSI Touchscreen

Official 7" DSI display + touch overlay for main dash UI.

MV07 — Video Streaming (WiFi AP)

RTSP server for camera feeds. View on phone/tablet connected to Pi AP.

MV08 — GStreamer Pipeline

Custom GStreamer for mux, encode, segment, and encrypt in one pipeline.

6.12 Body (10 integrations)

B01 — Door Open/Close Logger

Tap door jamb switches. Log each event with timestamp and GPS.

B02 — Hood / Trunk Status

Magnetic reed switches. Alert if open while driving or parked.

B03 — Seat Occupancy Sensor

Pressure mat or factory occupancy sensor tap. Enable passenger airbag logic display.

B04 — Window Position Estimator

Time-based estimation from window switch current sense. Alert if rain detected.

B05 — Sunroof Position

Hall effect or limit switch tap. Auto-close if rain + parked.

B06 — Mirror Position Memory

Tap mirror adjust motor wires. Save/recall positions per driver profile.

B07 — Wiper Speed Logger

Tap wiper motor low/high wires. Correlate with rain sensor (if added).

B08 — Washer Fluid Level

Float switch or capacitive sensor in reservoir. Alert if low.

B09 — Odometer Sync

Log GPS distance + OBD distance. Detect odometer discrepancy (tamper check).

B10 — Paint / UV Exposure Logger

UV sensor on exterior. Log cumulative exposure for paint care advice.

6.13 Fuel (8 integrations)

F01 — Fuel Level Monitor

OBD fuel level % (if supported) or analog tap on sender unit. Log with GPS.

F02 — Fuel Economy Real-Time

Instant and average MPG from MAF or injector duty cycle.

F03 — Fuel Station Finder

POI DB + current fuel level. Suggest station before empty.

F04 — Fuel Price Tracker

Manual entry + GasBuddy API (LTE). Log cheapest stations on route.

F05 — Ethanol Content Estimator

If flex-fuel sensor not present, estimate from fuel trim and O2 response.

F06 — Fuel Theft Detection

Sudden fuel level drop while parked + no ignition. Alert via LTE.

F07 — Range Estimator

Compute remaining range from fuel level / recent MPG. Display on dash.

F08 — Fuel System Leak Detection

Monitor fuel trim long-term. Large positive LTFT may indicate vacuum leak (acts like lean condition).

6.14 Communication (12 integrations)

COM01 — 4G/LTE Internet

Quectel EC25 via USB. wwan0 interface. NetworkManager for auto-connect.

COM02 — WiFi Hotspot

hostapd on Pi 5. Share LTE to passengers. WPA3 if supported.

COM03 — SMS Gateway

Receive commands via SMS. Send alerts. gammu or custom AT parser.

COM04 — GPS NMEA Stream

gpsd on /dev/ttyUSB0. Share to all apps via localhost:2947.

COM05 — Bluetooth OBD

ELM327 Bluetooth → rfcomm → python-obd. Wireless OBD option.

COM06 — ADS-B Receiver

RTL-SDR + dump1090. Display nearby aircraft on map. log to local DB.

COM07 — AIS Marine Receiver

RTL-SDR + rtl_ais. If near coast, log vessel traffic.

COM08 — LoRa Mesh (Optional)

SX1262 HAT. Off-grid text comms with other vehicles / base station.

COM09 — MQTT Telemetry

Publish sensor data to home MQTT broker (over LTE). Home Assistant integration.

COM10 — Remote SSH Access

Reverse SSH tunnel or Tailscale. Administer Pi from anywhere.

COM11 — Over-the-Air Updates

A/B partition or rsync-based OTA. Download on WiFi; apply on next boot.

COM12 — Emergency Beacon

On crash + no cell: cycle through bands (LTE → 2G → LoRa) to send SOS.

7. Wiring Diagrams

7.1 Overall System Block Diagram

```
+-----------------+       +-----------------+       +-----------------+
|  1996 Camry     |       |   12V→5V Buck    |       |  Raspberry Pi 5  |
|  Battery 12V    |------->|   Converter 5A   |------->|  USB-C Power     |
|  Fusebox        |       |  (TVS protected) |       |  (Active Cooler) |
```

```
+-----------------+       +-----------------+       +--------+--------+
       |                                                     |
       | Ignition Switched 12V                               | PCIe FPC
       v                                                     v
+-----------------+                                 +-----------------+
|  Relay Board    |                                 |  M.2 NVMe HAT   |
|  (4CH, 5V coil) |<--- GPIO17/27/22/23 ----------------->|  Samsung 990 EVO |
+-----------------+                                 |  1TB            |
       |                                            +-----------------+
       | Relay contacts                                    |
       v                                                   | CSI-0 / CSI-1
+-----------------+                                 +----v------------+
|  Accessory Power |                                |  Cameras (×2)   |
```

```
|  (Cameras, Hub)  |                                        |  IMX219 8MP      |
+------------------+                                        +------------------+
       |                                                           |
       | 12V always-on                                               | USB 3.0
       v                                                           v
+------------------+                                        +------------------+
|  LTE Modem       |<--- USB Hub ---------------------------->|  USB Hub (7pt)   |
|  (Quectel EC25)  |<--- USB -------------------------------->|  (Self-powered)  |
+------------------+                                        +------------------+
       |                                                           |
       | UART / I2C / SPI                                            | USB
       v                                                           v
+------------------+                                        +------------------+
```

```
|  GPS (u-blox)    |                                         |  RTL-SDR (×2)    |
|  MPU-6050 (I2C)  |                                         |  OBD ELM327      |
|  BME280 (I2C)    |                                         |  USB Audio       |
|  MCP2515 (SPI)   |                                         |  Coral TPU (opt) |
+------------------+                                         +------------------+
```

7.2 Ignition & Relay Wiring Detail

```
   Camry Fusebox (ACC/IG1)  ──→  Fuse 10A  ──→  Diode (1N5408)  ──→  Relay K1 coil+
                                                                     |
   Camry Chassis GND  ───────────────────────────────────────────────→  Relay K1 coil-
                                                                     |
   Relay K1 NO  ──→  Buck converter 12V IN+ (switched main power) |
   Relay K1 NC  ──→  (not used)                                     |
   Relay K1 COM ──→  Battery 12V (fused 15A)                      |
```

```
                                                          |
   Pi GPIO17 ──→  220Ω ──→  Optocoupler LED+ ──→  GND        |
   Optocoupler OUT ──→  Relay K1 coil+ (isolated)            |
                                                          |
   Pi GPIO27 ──→  220Ω ──→  Relay K2 coil+ (accessory cut)   |
   Relay K2 NO ──→  5V rail to USB Hub / Cameras             |
```

7.3 I2C Sensor Bus

```
   Pi GPIO2 (SDA1) ──┬──→ BME280 SDA (0x76 or 0x77)
                     ├──→ MPU-6050 SDA (0x68 or 0x69)
                     ├──→ OLED SSD1306 SDA (0x3C)
                     └──→ INA219 SDA (0x40)

   Pi GPIO3 (SCL1) ──┬──→ BME280 SCL
```

```
                       ├──→ MPU-6050 SCL
                       ├──→ OLED SCL
                       └──→ INA219 SCL


    All devices share 3.3V from Pi Pin 1 and GND from Pin 6.
    Add 4.7kΩ pull-ups to 3.3V if not present on modules.
```

7.4 OBD-II Connection (1996 Camry)

```
    Camry OBD-II port (driver knee panel, J1962 16-pin):


    Pin 4 (Chassis GND)  ───────────────────────────→  ELM327 GND
    Pin 5 (Signal GND)   ───────────────────────────→  ELM327 GND
    Pin 16 (Battery +)   ───────────────────────────→  ELM327 VCC (or not used if USB powered)
    Pin 2 (J1850 VPW)    ──X──  1996 Camry uses ISO 9141-2 or K-line (Pin 7)
    Pin 7 (K-Line)       ───────────────────────────→  ELM327 Pin 7 (ISO 9141-2)
```

```
Pin 15 (L-Line)      ──────────────────────→  ELM327 Pin 15 (optional)

ELM327 USB  ──→  Pi USB Hub (or direct)
```

> **Warning:** Do NOT short Pin 16 to ground. Do NOT connect to airbag pins (if present on non-standard connectors). Verify pinout with multimeter before permanent installation.

8. Installation Steps

Phase 1 — Preparation

1. Disconnect Camry negative battery terminal. Wait 10 minutes for capacitors to discharge.
2. Install Raspberry Pi OS (64-bit) on NVMe SSD or microSD. Enable SSH, I2C, SPI, UART via `raspi-config`.
3. Update firmware: `sudo rpi-eeprom-update -a` and `sudo apt update && sudo apt full-upgrade`.
4. Install NVMe HAT and SSD. Verify with `lsblk`.

5. Install active cooler. Verify thermal with `vcgencmd measure_temp` under load.

Phase 2 — Bench Test

1. Connect Pi to lab bench power supply (5V/5A). Boot and verify OS.
2. Connect cameras via CSI cables. Test with `rpicam-hello`.
3. Connect OBD-II adapter to a compatible vehicle or OBD simulator. Test with `python-obd` or `obd` library.
4. Connect GPS via USB or UART. Verify with `cgps` or `gpsmon`.
5. Wire I2C sensors on breadboard. Verify addresses with `i2cdetect -y 1`.
6. Test relay board with GPIO script. Verify switching with multimeter.

Phase 3 — Vehicle Install

1. Choose mounting location: behind glovebox, under driver dash, or in center console. Ensure airflow for Pi cooler.

2. Tap always-on 12V from fusebox (BAT circuit) with add-a-circuit fuse tap. Use 15A fuse.
3. Tap ignition-switched 12V from IG1 or ACC fuse. Use 10A fuse.
4. Mount buck converter in well-ventilated area. Secure with zip ties or screws. Ensure TVS diode installed.
5. Run fused 12V to buck input. Run buck 5V output to Pi USB-C via right-angle cable.
6. Mount relay board near Pi. Connect coil VCC to Pi 5V (or buck 5V). Connect coil GND to common ground.
7. Connect relay contacts to accessory loads (USB hub, cameras, LTE modem) per wiring diagram.
8. Route CSI cables from Pi to front and rear cameras. Avoid ignition coil and injector wiring. Use shielded extenders if >30cm.
9. Mount front camera behind windshield (legal placement). Mount rear camera inside rear window.

10. Mount GPS antenna on dash with clear sky view. Avoid metallic film windshields.

11. Connect OBD-II adapter and route USB cable to Pi. Secure adapter so it does not obstruct pedals.

12. Install LTE modem antenna on rear shelf or window. Keep away from GPS antenna.

13. Run I2C sensor wires inside dash. Secure with cable clips. Keep away from high-current wiring.

Phase 4 — Software Configuration

1. Enable `dtparam=pciex1` and `dtparam=nvme` in `/boot/firmware/config.txt`.

2. Configure `systemd` services for auto-start: data logger, video recorder, GPS logger, OBD poller.

3. Set up encrypted video partition: `gocryptfs /mnt/video_plain /mnt/video` or LUKS.

4. Configure network: WiFi for home sync, LTE for on-road. Use NetworkManager dispatcher scripts.

5. Install and configure `libcamera-apps`, `ffmpeg`, `python-obd`, `gpsd`.
6. Install local LLM: `llama.cpp` with `Llama-3-8B-Instruct-Q4_K_M.gguf`.
7. Configure voice pipeline: `whisper.cpp` → `llama.cpp` → `piper-tts`.
8. Set up MQTT client to publish telemetry. Configure Home Assistant auto-discovery if desired.
9. Write GPIO monitoring script for ignition sense → graceful shutdown timer.
10. Configure log rotation and NVMe wear monitoring: `smartctl -a /dev/nvme0`.

Phase 5 — Testing & Tuning

1. Start engine. Verify Pi boots and all services start. Check `dmesg` for errors.
2. Monitor 12V rail with INA219. Verify buck output is stable at 5.0–5.1V under load.
3. Test video recording: 30 min continuous. Check for thermal throttling. Adjust fan curve if needed.

4. Test OBD data stream at full rate. Verify no frame drops or bus errors.

5. Test ignition-off sequence: verify 10-min timer, graceful shutdown, relay cutoff.

6. Test LTE connectivity while driving. Verify handover between WiFi (home) and LTE (road).

7. Test emergency crash sequence: simulate high G with MPU-6050. Verify video save + SMS.

8. Test voice assistant in cabin noise. Adjust mic gain and noise gate.

9. Secure all wiring with loom, tape, and cable ties. No loose wires near pedals or steering.

10. Reconnect battery negative terminal. Perform final multi-meter check for shorts.

> **Final Safety Check:** Verify no wires interfere with brake pedal, accelerator, or steering column. Verify all fuses are correct ratings. Verify buck converter and Pi enclosure are not in airbag deployment zone. Document all non-factory wiring for future owners.

Appendix A — 1996 Toyota Camry Specific Notes

- **Engine:** 2.2L 5S-FE I4 or 3.0L 1MZ-FE V6. OBD-II protocol is ISO 9141-2 (K-line on Pin 7). Some early 1996 models may use J1850 VPW; verify with scan tool.

- **Transmission:** A140E (4-cyl) or A540E (V6) 4-speed automatic. No native CAN bus on this generation.
- **Fusebox:** Driver side kick panel and engine compartment. Use add-a-circuit taps for clean installation.
- **Ignition:** Switched 12V available at cigarette lighter (ACC) or ignition coil (IG1). Verify with multimeter.
- **Radio:** Standard double-DIN opening. Pi + 7" touchscreen can replace factory head unit with adapter kit.
- **Airbags:** SRS present. Never tap into SRS wiring. Keep all added wiring >30cm from airbag modules.

Appendix B — Quick Command Reference

```
# Check Pi temperature
vcgencmd measure_temp
```

```
# List I2C devices
i2cdetect -y 1

# Test camera
rpicam-hello --qt-preview

# Record H.264 segment
rpicam-vid -t 0 --inline --codec h264 -o test.h264

# OBD read DTCs (python)
import obd; connection = obd.OBD(); print(connection.query(obd.commands.GET_DTC))

# GPS status
cgps -s
```

```
# Check NVMe
sudo nvme smart-log /dev/nvme0

# Encrypt video partition
gocryptfs -init /mnt/video_cipher
gocryptfs /mnt/video_cipher /mnt/video
```

Appendix C — Integration Count Summary

Category	Count
Engine	15
Safety	12
Navigation	10
Infotainment	12
Climate	8
Security	12
Lighting	10
Data Logging	15
AI / LLM	10

Power Management	10
M.2 / Video	8
Body	10
Fuel	8
Communication	12
Total	**150**

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